Covariance & Correlation

Covariance measure how two variables change together.

- if both increase together — covariance is positive
- if one increase while the other decrease Covariance is negative
- if no relationship → covariance ≈ 0

Formula.

$$cov(X,Y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

where x_i, y_i = data points

$\bar{x}, \bar{y}$ = mean of x and y

n = number of data points.

Correlation :-

Correlation is a standard form of covariance. it tell us strength and direction of the relationship between two variables

(Pearson correlation Coefficient)

$$\rho = \frac{Cov(X,Y)}{\sigma_X \cdot \sigma_Y}$$

where σ_X = Standard deviation of X

σ_Y = Standard deviation of Y.

Range of γ

+1 ⇒ perfect positive correlation

-1 ⇒ perfect negative correlation

0 ⇒ No correlation

Key Insight :-

- Covariance : Tells the direction of relationship (positive or negative), but not strength.
- Correlation Tells both strength and direction in a standardized way (-1 & +1)

Applications

① Marketing and Sales :-

→ measure how ad spend correlates with sales. prioritize channels with stronger correlations

② Finance.

→ Correlation with market returns

③ manufacturing & quality

→ covariance b/w process input (temp. pressure and defect rates

(5) education Analytics :-

→ study-time V/s performance

attendance Vs grades;

identify meaningful patterns.

(6) Economic & policy.

interest rates vs investment, unemployment vs inflation.

(7) Data science preprocessing :-

→ feature selection (drop near duplicate feature with $r \approx 1$)

→ multicollinearity checks before regression.

(use correlation matrix)

multicollinearity happens when two or more independant variable (predictors) in a regression model are highly correlated with each other.

→ this means the predictors carry similar information.

As, as result it becomes hard for the model to figure out the individual effect of each predictor on the dependent variable..

Study Hours (x)	Exam Score (y)
2	40
4	50
6	60
8	70
10	80

Step ① calculate mean:-

$$\bar{x} = \frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$$

$$\bar{y} = \frac{40+50+60+70+80}{5} = \frac{300}{5} = 60$$

X	Y	$X-\bar{X}$	$Y-\bar{Y}$	$(X-\bar{X})(Y-\bar{Y})$	$(X-\bar{X})^2$	$(Y-\bar{Y})^2$
2	40	-4	-20	80	16	400
4	50	-2	-10	20	4	100
6	60	0	0	0	0	0
8	70	+2	+10	20	4	100
10	80	+4	+20	80	16	400

③ Calculate Covariance.

$$Cov(X,Y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{n-1}$$

$$= \frac{80+20+0+20+80}{5-1}$$

$$= \frac{200}{4} = 50$$

Covariance = 50 (positive) → as study hours increase, exam score also increase.

Step ④ Calculate standard deviation.

$$\sigma_X = \sqrt{\frac{(X-\bar{X})^2}{\sigma X}} = \sqrt{\frac{16+4+0+4+16}{4}} = \sqrt{\frac{40}{4}}$$

$$= \sqrt{10} = 3.16$$

$$\sigma_y = \sqrt{\frac{(y-\bar{y})^2}{n-1}} = \sqrt{\frac{400+100+0+100+400}{4}}$$

$$= \sqrt{1000/4} = \sqrt{250} \approx 15.81$$

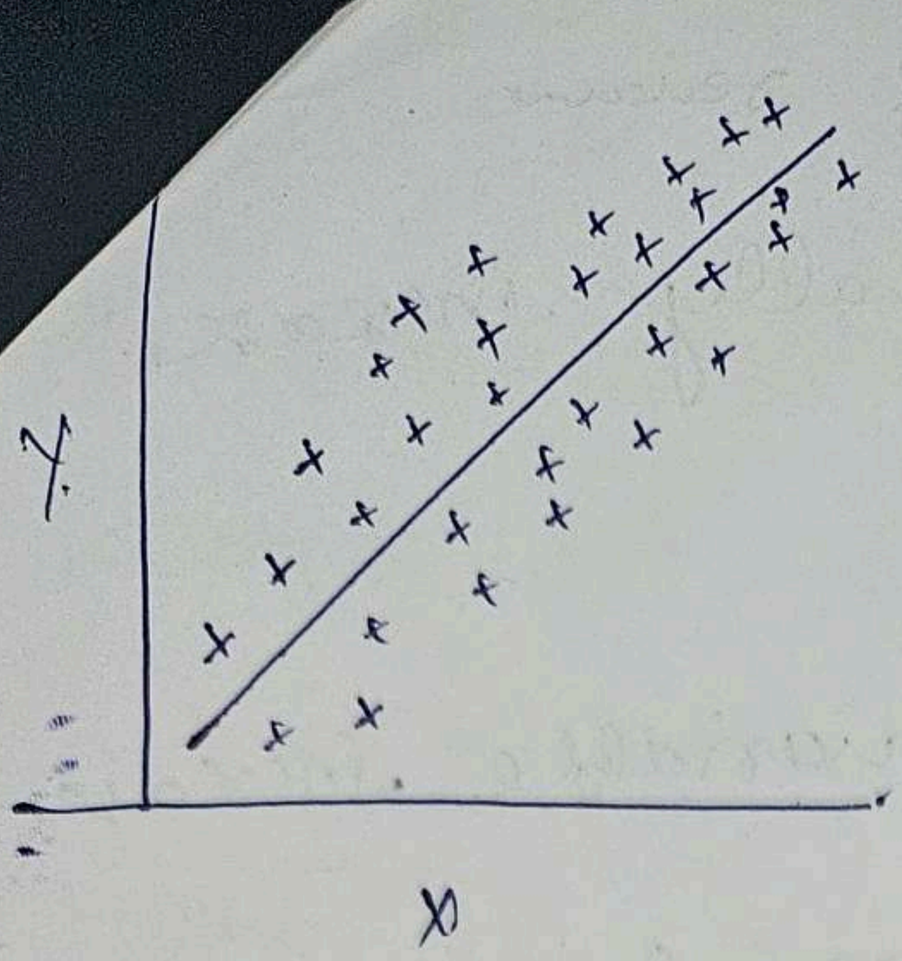
step (5) Calculate correlation coefficient:-

$$\gamma = \frac{Cov(X,Y)}{\sigma_x \cdot \sigma_y} = \frac{50}{(3.16)(15.81)}$$

$$= \frac{50}{50} \approx 1$$

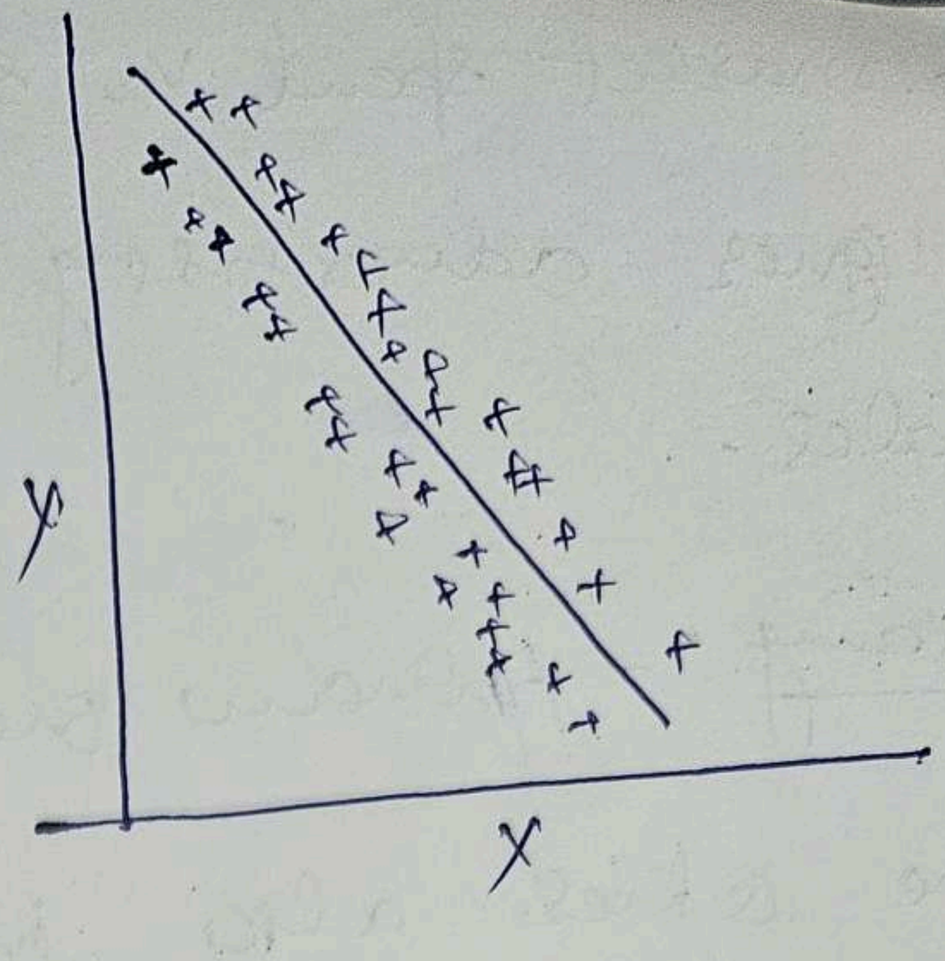
Correlation = +1 (perfect positive correlation)

→ More study hours directly lead to Higher exam score.



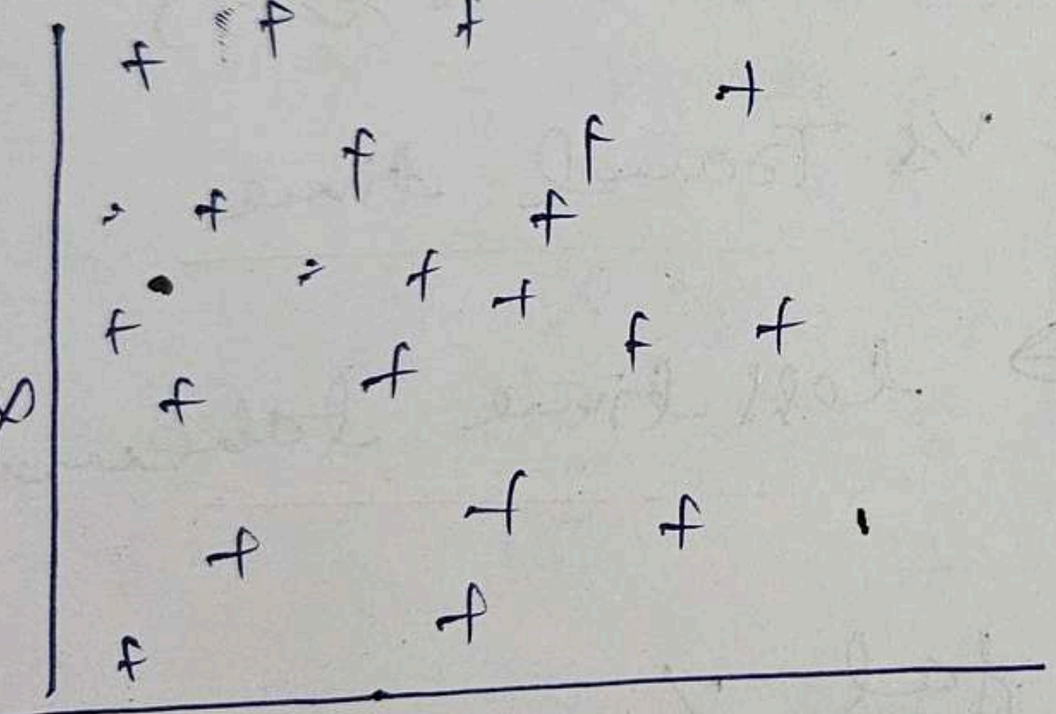
Strong positive correlation

(Cov > 0, r = +1)



Strong negative correlation

(Cov < 0, r = -1)



Strong positive Correlation (Cov > 0, r ≈ +1)

(No correlation

Cov ≈ 0, r ≈ 0)

=

① Study hours V/s Exam Score

The more a student studies, the higher their marks.

→ market spend vs sales revenue

Higher advertising usually increase sales.

meaning :- when one variable increase the other also increase consistently.

② Strong negative correlation (Cov < 0, $r \approx -1$)

→ Speed of a car vs Travel time

↑ Higher speed → less time taken ↓

→ Exercise vs body fat %.

more ↑ exercise → Lower body fat ↓.

→ Smoking vs Lung capacity

meaning :- when one variable increase the other decrease.

show size vs intelligence.

- favorite color vs annual income:

(No correlation)

- Hight vs Exam Score

- Car color vs Mileage (fuel efficiency)

Exp. X = Marketing spend. Y = monthly sales

Month	X (expenditure)	Y (sales)
		(in lac)
1	10	5
2	15	7
3	20	9
4	22	10
5	25	12
6	30	15
7	35	17
8	40	19
9	45	22
10	50	24

$$\text{mean}(\bar{x}) = \frac{\sum x_i}{n} = 29.2$$

$$\text{mean}(\bar{y}) = \frac{\sum y_i}{n} = 14.0$$

x	x	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$	$(x-\bar{x})^2$	$(x-\bar{x})^2$

$$\sum (x-\bar{x})(y-\bar{y}) = 762.0$$

$$\sum (x-\bar{x})^2 = 1557.6$$

$$\sum (y-\bar{y})^2 = 3740$$

number of points $n = 10$

(9) Sample variance

Sample covariance with $n-1$

$$Cov(X,Y) = \frac{\sum (X-\bar{X})(Y-\bar{Y})}{n-1}$$

$$= \frac{762.0}{9} = 84.666$$

interpretation: positive covariance

→ when marketing spend increases sales, sales tend to increase.

(5) Sample standard deviation

$$\sigma_x = \sqrt{\frac{(X-\bar{X})^2}{n-1}} = \sqrt{\frac{1557.6}{9}} = 13.155$$

$$\sigma_y = \sqrt{\frac{(Y-\bar{Y})^2}{n-1}} = \sqrt{\frac{374.0}{9}} = 6.4464$$

pearson correlation(r)

$$= \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{84.666}{(13.15)(6.44)}$$

$$= 0.99837$$

Interpret $r \approx 0.998$ → an extremely strong positive linear relationship (close to +1)

Covarience :- How do two variables change together.

Variance informs a researcher regarding the variability observed in a single variable while covariance provides information regarding the degree to which two variables vary together.

$$S_x^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}, \quad Cov.(X,Y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y - \bar{y})}{n-1}$$

the equation simply multiplies x's error score by y's error scores as opposed to squaring x's error scores.

Covariance :-

$$Cov(x,y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

- when $x \uparrow$ and $y \uparrow$: $cov(x,y)$ = positive.
- when $x \downarrow$ and $y \downarrow$: $cov(x,y)$ = negative.

when no constant relationship : $cov(x,y) = 0$